SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE,UNLIMITED POTENTIAL

DEPT OF CSE

AI-POWERED AIR QUALITY MONITORING

Real-Time AQI Intelligence for 20+ Indian Cities
Powered by Machine Learning Forecasting

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UNDER THE GUIDANCE OF

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THE AIR QUALITY CRISIS

Understanding pollution levels across 20+ Indian cities with real-time monitoring from CPCB stations



Current AQI Across Major Cities

AQI	City	Status	Trend	7d avg
312	Lucknow	SEVERE	Rising	298
298	Kanpur	SEVERE	Rising	285
287	Delhi	POOR	Rising	265
276	Patna	POOR	Falling	290
245	Jaipur	POOR	Rising	238
198	Kolkata	MODERATE	Falling	210
165	Ahmedabad	POOR	Rising	158
142	Mumbai	MODERATE	Falling	155
134	Hyderabad	MODERATE	Rising	128
112	Pune	MODERATE	Falling	125
96	Bangalore	SATISFACTORY	Falling	102
78	Chennai	SATISFACTORY	Rising	82

Data: CPCB Real-Time Monitoring Stations | Updated: May 2026

AI FORECASTING ENGINE

Machine Learning predictions powered by
Random Forest regression for 7-day forecasting



HEALTH INTELLIGENCE

Personalized health recommendations based on real-time AQI, age group, and medical conditions



Dynamic Health Advisory System

Actionable health advice updates dynamically based on current AQI readings

0-50 GOOD

No restrictions

Enjoy outdoor activities freely. Air quality is satisfactory with minimal health risk.

✓ Outdoor exercise
✓ Windows open
✓ No mask needed

51-100 SATISFACTORY

Normal activities

Minor precautions for sensitive individuals. Monitor symptoms if you have respiratory conditions.

✓ Most activities
△ Sensitive groups cautious
✓ No mask needed

101-200 MODERATE

Reduce exertion

Sensitive groups should limit outdoor exposure. Consider shorter outdoor sessions.

△ Limit prolonged outdoor
△ Sensitive groups stay in
△ Mask optional

201-300 POOR

Limit outdoor

Wear mask outside. Keep windows closed. Use air purifiers indoors if available.

✗ Avoid outdoor exercise
✗ Keep windows closed
● Wear N95 mask

300+ SEVERE

Stay indoors

Use air purifiers. N95 mask mandatory if going outside. Seek medical help if symptoms appear.

✗ Stay indoors
● N95 mask mandatory
● Air purifier essential

Personalized Protection by Age & Health

AGE-SPECIFIC RECOMMENDATIONS

Children

- Keep indoors when AQI > 150
- Limit outdoor play during high pollution
- Use N95 masks designed for kids
- Ensure schools have air purifiers

Seniors (60+)

- Avoid morning walks during high pollution
- Stay hydrated throughout the day
- Keep medications accessible
- Monitor for breathing difficulty

Pregnant Women

- Minimize outdoor exposure
- Use indoor air purification
- Avoid peak traffic hours
- Consult doctor if feeling unwell

Adults

- Follow standard AQI precautions
- Wear N95 when AQI > 200
- Limit outdoor exercise in poor air
- Use air purifiers at home/office

HEALTH CONDITION CONSIDERATIONS

Asthma

- Keep inhaler accessible at all times
- Monitor symptoms closely when AQI > 100
- Pre-medicate before going outside

Allergies

- Avoid peak pollution hours (morning/evening)
- Keep windows closed during high pollen + pollution
- Use antihistamines as prescribed

Heart Disease

- Minimize outdoor exertion when AQI > 150
- Seek medical advice if symptoms appear
- Avoid morning walks in poor air quality

PLATFORM FEATURES

Interactive visualization, city selection, and comprehensive data integration



FEATURES

KEY METRICS DISPLAYED



Current AQI with color-coded category badge



7-Day Average with trend direction indicator



Peak AQI from the last 30 days

Best AQI from the last 30 days



Daily Forecast for the next 5-7 days



Pollutant Breakdown with safe limit percentage bars


THANK YOU
Monitor. Forecast. Protect.